

REWEAR-FUSED Molecular Asset & Data-Sourcing Layer — Maximum-Depth Specification (June 2026)

TL;DR

- **Every molecule, structure, dataset, and ML model REWEAR-FUSED needs is downloadable, citable, and free for non-commercial demo use** — anchored on the real PDB depositions of the UMG-SP urethanase family (8WDW, 9FZW, 8S7Z, 9FVF) plus established α/β -hydrolase templates (6EQE, 7SH6, 6THS, 1TCA), with polymer chemistry drawn from PubChem CIDs for HDI/IPDI/PTMG/PCL, and polymer-property training data from PoLyInfo (552,427 data points as of 2025-05-29) + PI1M (~1M polymers) + the polyBERT 80 M-PSMILES training corpus.
 - **The static-artifact architecture is the credibility multiplier:** pre-compute .bcif (BinaryCIF) for proteins, .sdf for ligands, and JSON for properties/passports; serve over CDN with versioned filenames and SHA-256 checksums; render with Mol* (proteins) + 3Dmol.js / RDKit.js (small molecules) + React Three Fiber (polymer-chain narrative) — so the demo reads the same files an expert reviewer can re-download from rcsb.org or pubchem.ncbi.nlm.nih.gov.
 - **Be transparent about what is experimental vs predicted:** experimental crystal structures (UMG-SP2 8WDW, IsPETase 6EQE, etc.) are CC0; AlphaFold DB v6 predictions (~241 million proteins, CC-BY 4.0) and ESM Metagenomic Atlas (>700 million predicted structures in current v2023_02 per the facebookresearch/esm README; 617M was the original v0 figure from Lin et al. Science 379, 6637, 2023) are clearly labeled as predicted; Engine A and Engine B design outputs are explicitly tagged as "REWEAR-FUSED predicted, not experimentally validated." (Meta)
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Key Findings

1. **The UMG-SP urethanase family is fully PDB-anchored as of 2024-2025.** UMG-SP2 (PDB 8WDW, 2.59 Å apo + 2.16 Å PMS-bound + 2.40 Å BBC-bound, Cong/Li/Wei 2024; PDB 9FZW second deposit), UMG-SP1 (PDB 8S7Z, Bayer/Branson 2024 Angew Chem 202404492), and UMG-SP3 (PDB 9FVF, Rotilio 2025 Angew Chem 202419535, with two companion ligand-bound entries) give REWEAR-FUSED a real Ser-cisSer-Lys amidase-signature scaffold anchor. The Branson 2023 discovery paper (Angew Chem 62, e202216220) is the original source. (RCSB PDB + 2)
2. **PET hydrolases provide architectural Ser-His-Asp analogs** with deep precedent: IsPETase (6EQE, 0.92 Å; also 5XJH, 5XGO, 6ANE), ThermoPETase (6IJ6), DuraPETase (6KY5), FAST-PETase (7SH6), HotPETase (7QVH), wild-type LCC (4EB0), LCC-ICCG

(6THS S165A + 6THT active), MHETase (6QG9, 6QZ4). These are templates B can recombine without copying UMG-SP2 directly.

3. **RCSB PDB serves three structure file formats** (PDB legacy, PDBx/mmCIF, BinaryCIF/.bcif) via files.rcsb.org/download/{ID}.{ext} plus ModelServer (.bcif slices), with a GraphQL Data API at data.rcsb.org and a JSON Search API at search.rcsb.org. BinaryCIF is the production-grade format for web delivery — 3-10× smaller than mmCIF and natively consumed by Mol*. All PDB data is CC0.
4. **AlphaFold DB v6 (2025_03 UniProt sync) covers >241 million proteins** (+26M new vs v4-v5), accessed at https://alphafold.ebi.ac.uk/api/prediction/{UNIPROT_ID} returning JSON with download URLs for PDB/mmCIF/BinaryCIF/PAE-JSON; entry URL pattern alphafold.com/entry/AF-{ACCESSION}-F1; license CC-BY-4.0; per-residue pLDDT in B-factor column.
5. **ESM Metagenomic Atlas (Meta AI)** — the current v2023_02 release contains >700 million ESMFold-predicted metagenomic structures per the facebookresearch/esm GitHub README (CC-BY 4.0); the original 617M figure from Lin et al. Science 379, 6637 (2023) was the v0 release. Bulk-downloadable from [s3://dl.fbaipublicfiles.com/esmatlas/](https://s3.dl.fbaipublicfiles.com/esmatlas/), with a fold-on-demand API at <https://api.esmatlas.com/foldSequence/v1/pdb/>. Filter to pLDDT > 0.7, pTM > 0.7 for "interesting" structures. [GitHub + 3](#)
6. **PubChem PUG-REST is the workhorse for every small molecule:** pattern <https://pubchem.ncbi.nlm.nih.gov/rest/pug/compound/{namespace}/{id}/{operation}/{output}>. The pubchempy Python wrapper (v1.0.5) handles fetching SDF/JSON/PNG; license is public domain.
7. **For polymer notation, PSMILES (Ramprasad-Group/canonicalize_psmiles) is the de facto chemistry-tractable format;** BigSMILES (Olsen Lab MIT, github.com/olsenlabmit/BigSMILES, ACS Cent. Sci. 5, 1523, 2019) is the structurally rigorous stochastic format. Both are needed: PSMILES for ML, BigSMILES for the passport-grade polymer record. [ACS Publications](#)
8. **Polymer ML training data is now plentiful:** PI1M (~1M generated polymers, figshare DOI 10.6084/m9.figshare.12483473, Ma & Luo 2020); polyBERT (HuggingFace [kuelumbus/polyBERT](https://huggingface.co/kuelumbus/polyBERT), 600-D fingerprint) — per Kuenneth & Ramprasad, Nat. Commun. 14, 4099 (2023): "We use 80 million PSMILES strings for training and 20 million PSMILES strings for validation," drawn from a 100M corpus generated by BRICS reassembly of 13,766 known polymers into 4,424 fragments; PoLyInfo (NIMS, 19,227 homopolymers + 552,427 property data points as of 2025-05-29, including 67,704 Tg points and 22,652 tensile-strength points); Open Macromolecular Genome (synthetically-aware, OMG). [ACS Publications + 5](#)
9. **Carter's-grade garment realism comes from H&M Personalized Fashion Recommendations** (Kaggle, 105k articles + 1.37M customers, competition rules permit research use) for product metadata + Fashionpedia (48k images, 294 attributes, CC-BY-4.0)

for category ontology + GINETEX/ISO 3758:2023 for care symbols + GS1 GPC 8-digit Brick codes ([gpc-browser.gs1.org](https://www.gpc-browser.gs1.org)) for taxonomy.

10. **Chemical-compliance data flows from EPA CompTox Dashboard** (>1.37 million chemicals, DTXSID identifiers, CTX API requires free key from ccte_api@epa.gov, public domain) + EPA PFASMASTER list + PubChem hazard sections, featurized through ChemBERTa-2 (HuggingFace [DeepChem/ChemBERTa-77M-MTR](https://huggingface.co/DeepChem/ChemBERTa-77M-MTR)) and MoLFormer-XL (HuggingFace [ibm-research/MoLFormer-XL-both-10pct](https://huggingface.co/ibm-research/MoLFormer-XL-both-10pct)). (US EPA)
 11. **Environmental LCA layer** for the passport is anchored on the **EU PEFCR Apparel & Footwear v3.1** (published 29 April 2025; EC welcomed 25 June 2025; pefapparelandfootwear.eu; 16 EF 3.1 impact categories; durability multipliers; microplastic shedding module) plus the Higg MSI v3.11 (Cascale, 29 October 2025) for material-level scoring. (European Commission + 2)
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Details

PART 1 — Protein structures (Engine B + frontend protein rendering)

1a. RCSB PDB — the canonical source.

- Website: <https://www.rcsb.org>. Download endpoints (files.rcsb.org):
 - Legacy PDB: <https://files.rcsb.org/download/{ID}.pdb> (~100-500 kB per entry, single-model proteins)
 - PDBx/mmCIF: <https://files.rcsb.org/download/{ID}.cif> (~200 kB-2 MB, current archival format)
 - BinaryCIF: <https://models.rcsb.org/{ID}.bcif> (typically 30-200 kB; **use this for the web demo**)
 - Gzipped variants via [.gz](#) suffix; range requests supported.
 - ModelServer slices (single chain, ligand, residue range):
<https://models.rcsb.org/v1/{ID}/full?encoding=bcif>
- Data API (GraphQL + REST): <https://data.rcsb.org/rest/v1/core/entry/{ID}> returns JSON with title, citation, experimental method, resolution, R-factor, ligands.
- Search API (JSON DSL, POST to <https://search.rcsb.org/rcsbsearch/v2/query>) supports text, attribute, sequence (MMseqs2), structure (BioZernike) and motif queries; can include AlphaFold/ModelArchive CSMs via [results_content_type: \["experimental", "computational"\]](#). (Biotite-python)
- Python clients: [rcsbapi](#) (Piehl et al., J. Mol. Biol. 2025, DOI 10.1016/j.jmb.2025.168970), [biotite.database.rcsb](#), [pypdb](#).

- **License: CC0 1.0 Universal** for coordinate files. Rate-limit: ~1-5 req/s; static-file downloads are unthrottled.

1b. Specific REWEAR-FUSED-relevant structures (download list).

Role	PDB ID	Title	Resolution	Notes
UMG-SP2 apo (primary anchor)	8WDW	Crystal structure of novel PU plastic degradation urethanase UMG-SP2 (uncultured bacterium) RCSB PDB	2.59 Å (apo); 2.16 Å (PMS); 2.40 Å (BBC) Wiley Online Library	Cong, Li, Wei et al. 2024; Ser-cisSer-Lys triad; ResearchGate L3 loop 219-226 governs activity ResearchGate
UMG-SP2 (2nd deposit)	9FZW	Structure of Urethanase UMG-SP2 RCSB PDB	-	Rotilio et al. companion
UMG-SP1	8S7Z	Urethanase UMG-SP1 (flexible active-site loops) RCSB PDB	-	Bayer et al. 2024 Angew Chem 202404492 Wiley Online Library
UMG-SP3	9FVF	UMG-SP3 amidase from uncultured bacterium RCSB PDB	-	Rotilio 2025 Angew 202419535; three states: ligand-free, suicide-inhibitor, product-bound PubMed
Related: AtzE 1-carboxybiuret amidase	6C62 / 6C6G	AtzE + PPDI suicide inhibitor (Pseudomonas sp. ADP) Rcsb	-	Ser-cisSer-Lys
Hydrazidase (Microbacterium HM58-2)	5H6T (wt) / 5H6S (S179A + HBH)	Hydrazidase	-	Ser-cisSer-Lys; geometric analog
ClbL amidase (E. coli, colibactin BGC)	8ES6	ClbL bound to PMS RCSB PDB	1.9 Å RCSB PDB	Ser-cisSer-Lys IUCr PubMed (S171/S155/K80) ResearchGate
NylA nylonase A	3A2P / 3A2Q	Nylonase A WT / S174A + Acd RCSB PDB	1.8 Å RCSB PDB	Arthrobacter

Role	PDB ID	Title	Resolution	Notes
Maleamate amidase Ami	4L08	Ami(C149A) + maleate (P. putida S16) RCSB PDB	-	AS-family carbamate analog
PET hydrolase (Ser-His-Asp analog)	6EQE	IsPETase, I. sakaiensis	0.92 Å	Han et al.; gold-standard α/β -fold
IsPETase variants	5XJH, 5XG0, 6ANE	various WT RCSB PDB	1.5 Å RCSB PDB	Joo et al.
ThermoPETase	6IJ6	S121E/D186H/R280A PubMed Central	-	
DuraPETase	6KY5	GRAPE-designed; Tm ~77 °C ResearchGate	-	
FAST-PETase	7SH6	ThermoPETase + N233K/R224Q bioRxiv	-	Lu et al., Nature 2022
HotPETase	7QVH	21-mutation evolved IsPETase PubMed Central	-	Hardy et al., Nat Catal 2022
Wild-type LCC	4EB0	Leaf-branch compost cutinase	-	Sulaiman et al., released 2013-03-27 RCSB PDB
LCC-ICCG	6THS (S165A) / 6THT (active)	Tournier engineered LCC	high-res	Tournier et al., Nature 2020
MHETase	6QG9 / 6QZ4 / 6QGA / 6QGB / 6JTU	I. sakaiensis MHETase	1.6 Å RCSB PDB	Knott et al.

Role	PDB ID	Title	Resolution	Notes
CalB lipase	1TCA	Candida antarctica lipase B RCSB PDB	1.55 Å PubMed Central	Uppenberg et al.; canonical Ser-His-Asp RCSB PDB
Thermomyces lanuginosus lipase	1DT3 (+1DT5, 1DTE, 1DU4, 1EIN, 4ZGB, 4DYH, 4EA6, 4FLF, 4GI1, 4GLB)	TLL	-	
Humicola insolens cutinase	4OYY (apo) / 4OYL (MEP-Ser) RCSB PDB RCSB PDB	HiC	2.05 Å	

1c. AlphaFold Protein Structure Database.

- URL: <https://alphafold.ebi.ac.uk> (canonical) and <https://alphafold.com> (mirror). v6 released aligned with UniProt 2025_03 — over 241 million predicted structures (+26M new vs v4-v5). [EMBL-EBI](#)
- Entry URL: <https://alphafold.com/entry/AF-{UNIPROT}-F1> (e.g. AF-P40763-F1).
- API: <https://alphafold.ebi.ac.uk/api/prediction/{UNIPROT}> returns JSON listing pdbUrl, cifUrl, bcifUrl, paeImageUrl, paeDocUrl. Docs: <https://alphafold.com/api-docs>.
- File patterns: PDB at https://alphafold.ebi.ac.uk/files/AF-{UNIPROT}-F1-model_v6.pdb; mmCIF similarly; PAE at [...-predicted_aligned_error_v6.json](#).
- For Titin-length proteins (>2700 aa), fragments F1, F2 (residues 201-1600), etc.
- Per-residue pLDDT in B-factor column: >90 very high, 70-90 confident, 50-70 low, <50 very low.
- **License: CC-BY-4.0.** Cite Jumper et al. Nature 2021 and Bertoni et al. NAR 2025.
- Bulk: bulk per-organism tar.gz downloads, plus FTP at <https://ftp.ebi.ac.uk/pub/databases/alphafold/> (includes v4 archive). [EMBL-EBI](#)
- Python: [Bio.PDB.alphafold_db](#) (Biopython ≥1.85), [AlphaFoldDB Structure Extractor](#) (Saraf et al. BMC Bioinformatics 2025).

1d. ESM Metagenomic Atlas.

- URL: <https://esmatlas.com>. The current v2023_02 release holds **more than 700 million** ESMFold-predicted metagenomic structures per the facebookresearch/esm GitHub README (the 617M figure was the v0 release in Lin et al. Science 379, 6637, 2023). Bulk: s3://dl.fbaipublicfiles.com/esmatlas/v0/highquality_clust30/ (tarballs + FoldSeek DB). [GitHub](#)
- Fold-on-demand API: `curl -X POST --data "{SEQUENCE}" https://api.esmatlas.com/foldSequence/v1/pdb/`. [GitHub](#)
- High-confidence subset: pLDDT > 0.7 + pTM > 0.7; FASTA at https://dl.fbaipublicfiles.com/esmatlas/v0/highquality_clust30/highquality_clust30.fasta. [GitHub](#)
- **License: CC BY 4.0** (Meta Open Source). Use for sourcing additional carbamate-hydrolase candidates from metagenomic dark matter.

1e. UniProt.

- URL: <https://www.uniprot.org>. REST: <https://rest.uniprot.org/uniprotkb/{ACCESSION}.{fasta|json|xml}>.
- UMG-SP2 sequence retrievable via GenBank accession OP972510 (Branson et al.) — its UniProt assignment derives from the published genome. [ACS Publications](#)
- Used to feed sequences to RFdiffusion2 / LigandMPNN / Boltz-2.

1f. Mol (molstar) rendering recipe.*

- npm [molstar](#) (current 4.x stable; check npmjs.com/package/molstar). Companion [pdbe-molstar](#) (web component, used by AlphaFoldDB) and [rcsb-molstar](#) (used by RCSB).
- React: [molstar-react](#) (community wrapper) takes `pdbId="8WDW"` or `customData={url, format: 'pdb' | 'mmcif' | 'bcif'}`.
- URL-loadable viewer: <https://molstar.org/viewer/?structure-url=https://models.rcsb.org/8WDW.bcif&structure-url-format=mmcif&structure-url-is-binary=1>.
- Snapshot files ([.molj](#)) / ([.molx](#)) capture camera + representations; ideal to ship pre-baked publication-grade views.
- Catalytic-triad highlight pattern: load 8WDW → select chains and residues Ser190 + Lys/cisSer + carbamate substrate BBC → custom representation "licorice" colored by element + cartoon dim + surface translucent.
- For Ser-His-Asp triad on PET hydrolase analog (6EQE): Ser160, Asp206, His237 shown as sticks; cartoon yellow β-strands + purple α-helices.

1g. Publication-grade complex visualization.

PART 2 — Small-molecule, ligand & chemical structure sources

2a. PubChem PUG-REST.

- Base: `(https://pubchem.ncbi.nlm.nih.gov/rest/pug)`.
- Patterns:
 - By CID → SDF (2D): `(/compound/cid/{CID}/record/SDF)`
 - 3D conformer: `(/compound/cid/{CID}/record/SDF?record_type=3d)` `(Iupac)`
 - SMILES: `(/compound/cid/{CID}/property/IsomericSMILES/TXT)`
 - Properties batch: `(/compound/cid/{CID1}, {CID2}/property/MolecularWeight,XLogP,TPSA/CSV)`
 - Hazard view: `(/pug_view/data/compound/{CID}/JSON?heading=GHS%20Classification)`
- Python: `(pubchempy)` 1.0.5 `((pcp.download('SDF', name, 'name', 'aspirin.sdf')))`.
- **License: PubChem data is public domain.** Aim for ≤5 req/s; HTTP POST for long SMILES with `(/)` characters.

2b. RCSB Chemical Component Dictionary / Ligand Expo.

- URL: `(https://www.rcsb.org/ligand)` (3-letter chem-comp IDs; for UMG-SP2: "BBC" benzyl carbamate substrate, "PMS" phenylmethanesulfonyl suicide inhibitor).
- Download CCD entries: `(https://files.rcsb.org/ligands/download/{CCD_ID}.cif)` (or `(.sdf)`).
- Build the carbamate-bond cartoon by extracting BBC from 8WDW with ModelServer ligand-mode → SDF.

2c. Other small-molecule sources.

- **ChEMBL** (EMBL-EBI; `(https://www.ebi.ac.uk/chembl)`; v34, released 28 March 2024 per the EMBL-EBI ChEMBL team blog post "ChEMBL 34 is out!" — contains exactly **2,431,025 compounds** (of which 2,409,270 have mol files); REST + SQLite dumps; CC-BY-SA 3.0) — for bioactivity if Engine B needs binding precedents.
- **ChemSpider** (RSC; `(https://www.chemspider.com)`; 100M+ structures; free reads, API key needed).

- **Cambridge Structural Database (CSD)** (CCDC; ~1.3M small-molecule crystal structures; commercial license, but the Teaching Subset and free CSD-Community access cover demo needs; the ETKDG torsion library inside RDKit was trained on CSD data).
- **ZINC22** (UCSF Shoichet Lab; <https://cartblanche.docking.org/>; **37.2 billion** purchasable molecules per Tingle et al., J. Chem. Inf. Model. 2023, DOI 10.1021/acs.jcim.2c01253; the tranche browser organizes them "by heavy atom count and lipophilicity"; free). Wikipedia

2d. Polymer representation.

- **SMILES** (Weininger 1988) — small molecule baseline.
- **PSMILES** (Polymer SMILES, Ramprasad Group) — repeat-unit SMILES with `[*]` endpoints; canonicalize via github.com/Ramprasad-Group/canonicalize_psmiles (pip install psmiles).
- **BigSMILES** (Olsen Lab MIT, ACS Cent. Sci. 2019, 5, 1523) — full stochastic representation with `{ }` blocks and `<>` bonding descriptors; tools at github.com/olsenlabmit/BigSMILES (BigSMILES Builder GUI + PolyDAT data schema). A cleavable segmented PU-urea looks like `{ [<] [>] C(=O)NRNC(=O)O-PolySoft-O [<] [>] }`.
- **3D conformer generation:** RDKit (current 2026.03.2 stable) `AllChem.EmbedMolecule(mol, AllChem.ETKDGv3())`; ETKDGv3 default since 2024.03 release, uses CSD torsion preferences; macrocycle-aware. Build the polymer chain by concatenating repeat units (e.g. 3-5 repeats) and embedding the linear fragment, optionally minimizing with MMFF/UFF. RDKit RDKit
- **Segmented block visualization:** Render hard segment (urea-rich, e.g. HDI + ethylenediamine extender) + soft segment (PTMG or PCL diol) in distinct colors via 3Dmol.js or React Three Fiber, with the carbamate -NH-C(=O)-O- bond highlighted as the cleavage site.

2e. Exact chemistry for the cleavable elastane.

Component	Role	PubChem CID	SMILES
HDI (hexamethylene diisocyanate)	aliphatic diisocyanate	13242	<chem>O=C=NCCCCCCN=C=O</chem>
IPDI (isophorone diisocyanate)	aliphatic diisocyanate	17394	<chem>O=C=NCC1(C)CCC(C)(N=C=O)CC1</chem>
H12MDI (4,4'-methylenebis(cyclohexyl isocyanate))	aliphatic diisocyanate	16156	<chem>O=C=NC1CCC(CC1)CC1CCC(CC1)N=C=O</chem>
PTMG (poly(tetramethylene ether glycol), Mn≈2000)	soft segment	repeat	<chem>[*]OCCCC[*]</chem> -
PCL diol (polycaprolactone diol)	soft segment	repeat	<chem>[*]OC(=O)CCCC[*]</chem> -
Polycarbonate diol (HDO-based)	soft segment	repeat	<chem>[*]OC(=O)O(CH2)6[*]</chem> -
1,4-Butanediol (BDO)	chain extender	8064	<chem>OCCCCO</chem>
Ethylenediamine (EDA)	urea-chain extender	3301	<chem>NCCN</chem>
Carbamate linkage	cleavage site	-	<chem>R-NH-C(=O)-O-R'</chem>

2f. Web visualization toolkit.

Library	npm	Latest (2026)	Best for	License
Mol* (molstar)	<code>molstar</code>	4.x	Proteins, electron density, large biomol assemblies	MIT
pdbe-molstar	<code>pdbe-molstar</code>	3.x	Web-component embedding with AlphaFold DB conventions	Apache 2.0
3Dmol.js	<code>3dmol</code>	2.5.5 <code>npm</code> (Dec 2025)	3D small molecules + proteins, WebGL, <code>npm</code> simple API	BSD-3-Clause
RDKit.js	<code>@rdkit/rdkit</code>	2025.3.4-1.0.0 <code>npm</code> (Jul 2025)	2D depictions, SMILES parsing, descriptors, WASM	BSD-3-Clause
SmilesDrawer	<code>smiles-drawer</code>	2.3.0 <code>npm</code> (May 2026)	Fast 2D SVG/canvas depictions, zero deps	MIT
Kekule.js	<code>kekule</code>	1.x	Chemoinformatics toolkit (read/write/edit/algorithms)	MIT
Ketcher	<code>ketcher-react</code> + <code>ketcher-core</code>	3.12.0 <code>npm</code> (Apr 2026)	Full 2D structure editor, macromolecules supported	Apache-2.0
React Three Fiber	<code>@react-three/fiber</code>	8.x	Custom narrative 3D scenes (polymer chain ribbon, cleavage animation)	MIT

PART 3 — Polymer property & materials datasets (Engine A)

3a. Polymer-property databases.

- NIMS PoLyInfo** — <https://polymer.nims.go.jp>. As of **2025-05-29**: 19,227 homopolymers; 8,321 copolymers; 2,788 blends; 3,209 composites; 174,968 samples; **552,427 property data points**; 21,793 literature records. Property coverage: Tg 67,704 points, Tm 42,543, tensile stress-at-break 22,652, density 14,674, gas permeability 13,981, dielectric constant 7,923. Access: DICE account registration required; **web scraping prohibited**; programmatic bulk download not offered (must request). `nims` `nims`
- PI1M** — Ma & Luo, J. Chem. Inf. Model. 2020, DOI 10.1021/acs.jcim.0c00726; figshare DOI 10.6084/m9.figshare.12483473; GitHub github.com/RUIMINMA1996/PI1M. ~1M generated

polymers in PSMILES; RNN trained on ~12,000 PoLyInfo seeds. CSV format. License: open for research. (ACS Publications)

- **polyBERT** — Kuenneth & Ramprasad, Nat. Commun. 14, 4099 (2023). The paper states: "We use 80 million PSMILES strings for training and 20 million PSMILES strings for validation," drawn from a 100M corpus generated by BRICS-reassembly of 13,766 known polymers into 4,424 fragments. Model on HuggingFace: (kuelumbus/polyBERT) (600-D fingerprint output, sentence-transformers compatible). Academic Research Use License (non-commercial only). Zenodo: 10.5281/zenodo.7969082. (Phys.org + 2)
- **Khazana** — <https://khazana.gatech.edu> (originally khazana.uconn.edu). Huan et al., Sci. Data 3, 160012 (2016); 1,073 DFT-uniform polymer entries initially, now expanded; computed band gaps, atomization energies, dielectric constants. Open. (Gatech + 2)
- **Polymer Genome** — <https://www.polymergenome.org>. Web service (no bulk download) for property prediction; underlying datasets sourced from Khazana + PoLyInfo + group-internal. Free academic use; commercial via Matmerize. (ADS) (ACS Publications)
- **Open Macromolecular Genome (OMG)** — Kim et al., ACS Polym. Au 2023; synthetically-aware database tying CRUs to monomer reactants + reaction pathways; ML-ready VAE (Molecule Chef) for synthesizable generation. Open.

3b. Polyurethane-specific datasets.

- **Ding 2023 XGBoost PU dataset** (Macromolecules 56, 8055): hundreds of curated PU mechanical data points; supplementary CSV available.
- **Xu 2026 SusMat PU LCA + property data**: published with article supplementary.
- **Qin 2024** (search-driven PU design): supplementary CSV with Tg, modulus, elongation.
- Manual augmentation from PoLyInfo subsetting on PU class.

3c. Elastane/spandex reference values (literature-cited for the trade-off curve):

- Soft segment Tg: PTMG ≈ -84 °C; PCL ≈ -60 °C.
- Hard segment Tg: HDI-urea $\approx 80-100$ °C, MDI-urea $\approx 110-140$ °C.
- Typical elongation at break: 400-800% for elastane.
- Hard-segment wt%: 5-30%; <10% loses recovery, >30% loses elasticity.
- Tensile strength: 50-150 MPa at break. Sources: van Krevelen "Properties of Polymers" 4th ed. (Elsevier 2009); Bicerano "Prediction of Polymer Properties" 3rd ed. (Marcel Dekker 2002); Hepburn "Polyurethane Elastomers" (Elsevier).

3d. Group-contribution fallbacks (for properties not predicted by polyBERT confidently): van Krevelen group-contribution for Tg, density, cohesive energy density; Bicerano connectivity-index method. Both formalize as Python functions over RDKit-parsed PSMILES.

PART 4 — Garment, product, textile & material-composition data (Engine C)

4a. Public apparel/product datasets.

- **H&M Personalized Fashion Recommendations** — <https://www.kaggle.com/competitions/h-and-m-personalized-fashion-recommendations>. 105k articles × 1.37M customers × 2 years transactions. CSVs: `articles.csv` (article_id, product_code, prod_name, product_type, garment_group, color, department, section, perceived_colour_value), `customers.csv`, `transactions_train.csv`. 30+ GB images. **Limitation: no fiber composition** — fiber composition must be augmented from category prior. `Medium` `Hugging Face`
- **Fashionpedia** — <https://fashionpedia.github.io>. 48k images + 27 categories + 19 parts + 294 fine-grained attributes (fabric, neckline, pattern, length, opacity, etc.). **CC-BY-4.0**. HuggingFace: `detection-datasets/fashionpedia`. `Hugging Face` `Hugging Face`
- **DeepFashion** — <http://mmlab.ie.cuhk.edu.hk/projects/DeepFashion.html>. 800k+ images, 50 categories, 1000 attributes. License: non-commercial research only; agreement required. `Cuhk` `Cuhk`
- **DeepFashion2** — 13 categories with instance segmentation.

4b. Children's/baby apparel.

- No purpose-built public dataset for Carter's-class baby apparel exists with full fiber composition; build synthetic-but-realistic records by:
 1. Taking H&M `garment_group` = "Baby/Children" rows.
 2. Augmenting fiber composition using category priors (e.g., baby onesie ≈ 95% cotton / 5% elastane; sleeper ≈ 60% cotton / 38% polyester / 2% elastane; legging ≈ 90% cotton / 10% elastane).
 3. Validating against published industry blend ratios. Textile Exchange Materials Market Report 2024 reports cotton fell from 25.1 Mt (2022) to 24.4 Mt (2023) against a total global fiber production of 124 Mt (up 7% from 116 Mt in 2022) — i.e. cotton was 19.7% of global fiber in 2023; elastane volumes are smaller globally (~1%) but present in 30-60% of stretch knit garments. `Cascale + 2`

4c. Fiber reference data + GS1 GPC.

- **GS1 GPC Browser** — <https://gpc-browser.gs1.org>. Per GS1 documentation: "only the 8-digit 'Brick' code is accepted. These codes will always start with '1000'." Hierarchy: Segment → Family → Class → Brick → Brick Attribute. Apparel maps to the Clothing Segment. Downloads: JSON / XML / XLSX / "Combined Published Schema" zip. `Translate + 2`

- **Textile fiber priors:** Textile Exchange Materials Market Report 2024 (<https://textileexchange.org/publications/>) for global fiber prevalence.

4d. Care-label and labeling standards.

- **ISO 3758:2023** — <https://www.iso.org/standard/74401.html>. 4th edition (Dec 2023), replaces ISO 3758:2012. Defines 5 main symbols (washing, bleaching, drying, ironing, professional care) + variations. Added "wash by hand at ambient temperature" symbol; harmonized with GINETEX symbols (which co-own trademarks). (SGS + 2)
- **GINETEX** — <https://www.ginetex.net/gb/labelling/care-symbols.asp>; symbols are trademarked and require licensing for commercial use but reference/demo OK. (Ginetex)
- **EU Regulation 1007/2011** — fiber names + composition labeling.
- **US FTC Textile Fiber Products Identification Act** — <https://www.ftc.gov/legal-library/browse/statutes/textile-fiber-products-identification-act>.

4e. Lifecycle / environmental data.

- **Higg MSI v3.11** (Cascade, **29 October 2025**, <https://cascade.org/topic/higg-msi/>) — cradle-to-gate scoring; new cotton LCA methodology (Oct 2024) + updated nylon/polyester models + new home-textile materials (stoneware, glass ceramic, porcelain, container glass); exclusive hosting on Worldly; background data from Sphera Managed LCA Content (formerly GaBi). Programmatic access via Worldly API (paid). For free demo: cite Higg MSI baseline scores published in Cascade documentation. Cascade: "10 points per impact category represent the impact of the 'average material'." (Cascade + 6)
 - **EU PEF CR Apparel & Footwear v3.1** — published **29 April 2025**, EC welcomed **25 June 2025**; portal <https://pefapparelandfootwear.eu>; full PDF mirror https://www.carbonfact.com/hubfs/A&FW_PEF CR_v3.1.pdf. Defines functional unit, BoM by material share, intrinsic-durability multipliers, Circular Footprint Formula R2/R3, default duration-of-service, microplastic shedding module. 16 EF 3.1 impact categories. Disallows single-PEF-score B2C communication: "The PEF CR for apparel and footwear (version 3.1) does not allow the use of the PEF single score for business-to-consumer (B2C) communications... at least the four most relevant impact indicators." (European Commission)
 - **Ecoinvent v3.10** (<https://ecoinvent.org>, paid; some textile processes via academic access).
 - **Open LCA Nexus** — <https://nexus.openlca.org> — free textile datasets (Agribalyse, USDA, etc.).
 - **WRAP Textiles 2030 UK data** — open reports.
-

PART 5 — Chemical-compliance & safety (Engine C molecular layer)

5a. EPA CompTox Chemicals Dashboard + CTX APIs.

- URL: <https://comptox.epa.gov/dashboard>. As of latest release: **1,376,722 chemicals**. DSSTox identifier DTXSID format: "DTXSID" + 7-9 digits. (USEPA) (EPA)
- API access: free key from ccte_api@epa.gov; base URL (<https://api-ccte.epa.gov/>). Endpoints:
 - Chemical details: (</chemical/detail/search/by-dtxsid/{DTXSID}?projection=chemicaldetailstandard>)
 - List membership: (</chemical/list/>)
 - Hazard / ToxValDB: (</hazard/...>)
- **License: public domain** — per EPA: "These data are free of all copyright restrictions, and are fully and freely available for both non-commercial and commercial use." (USEPA)
- Bulk downloads: <https://www.epa.gov/comptox-tools/downloadable-computational-toxicology-data>.
- **PFAS Master List**: <https://comptox.epa.gov/dashboard/chemical-lists/pfasmaster> (consolidated PFAS list; sub-lists include EPAPFASINV ToxCast inventory, PFASOECD OECD New Global DB, PFASKEMI Swedish KEMI, PFASSTRUCT structure-based, PFASDEV1 polymer/UVCB).

5b. PubChem hazard view + ECHA.

- PubChem heading "GHS Classification" via PUG-View: (https://pubchem.ncbi.nlm.nih.gov/rest/pug_view/data/compound/{CID}/JSON?heading=GHS%20Classification).
- ECHA SVHC list: <https://echa.europa.eu/candidate-list-table>; CLP harmonized C&L: <https://echa.europa.eu/information-on-chemicals>.
- CMR (Carcinogenic/Mutagenic/Reprotoxic) classifications mappable via DTXSID ↔ CASRN ↔ EC number.

5c. Pretrained chemical language models.

- **ChemBERTa-2** — Ahmad et al., arXiv:2209.01712 (2022). HuggingFace ([DeepChem/ChemBERTa-77M-MTR](#)), ([DeepChem/ChemBERTa-77M-MLM](#)), ([DeepChem/ChemBERTa-10M-MTR](#)). Trained on 77M unique SMILES from PubChem (canonicalized). MLM + multi-task regression objectives. RoBERTa backbone. (arxiv + 2)
- **MoLFormer-XL** — Ross et al., Nat. Mach. Intell. 4, 1256 (2022). HuggingFace ([ibm-research/MoLFormer-XL-both-10pct](#)) (10% ZINC + 10% PubChem, ~110M molecules); also

`ibm-research/GP-MoLFormer-Uniq` for generation. Linear attention + rotary embeddings; ≤ 202 -token SMILES. `Hugging Face + 2`

- **Mol2vec** — github.com/samoturk/mol2vec; word2vec over Morgan substructures.
- **ChemBERTa-1** — Chithrananda et al., arXiv:2010.09885 (2020).
- **RDKit Morgan/ECFP** (radius 2, 2048 bits) — universal classical baseline.

5d. Mapping detected chemical → structure → hazard. Pipeline: detected name/CAS → DTXSID lookup via CTX API or DSSTox CSV → canonical SMILES → PubChem CID → GHS classification + ECHA SVHC flag → CASRN cross-ref → final hazard score. Featurize via ChemBERTa-2 or Morgan FP for ML-based hazard prediction.

PART 6 — Data infrastructure, provenance & static-artifact pipeline

6a. Static-artifact architecture.

- **File formats** (priority order for web):
 - Proteins: **BinaryCIF (.bcif)** primary (gzipped); .pdb fallback for legacy clients.
 - Ligands: SDF (3D), CIF (geometry from Ligand Expo).
 - Polymers: PDB or XYZ (RDKit-generated 3D), plus JSON metadata (PSMILES, BigSMILES, repeat-unit definitions).
 - Properties / passport / dataset rows: JSON (one file per garment/design/run).
 - ML model weights/embeddings: HF Hub or local Parquet.
- **Hosting:** Cloudflare R2 / AWS S3 + CloudFront / Vercel Blob / GitHub LFS for repo-coupled artifacts. Public, immutable, versioned filenames (e.g. `artifacts/v1.2.0/proteins/8WDW.bcif`).
- **Versioning + checksums:** SHA-256 per file in a top-level `manifest.json`; bump version on any structural change.
- **Bundle size:** a complete demo with 20 proteins + 50 ligands + 10 designed polymers + passport JSONs fits in ~5-15 MB compressed.

6b. Provenance + reproducibility.

- Every artifact carries:
 - Source database name + version
 - Accession ID (PDB 4-char, UniProt, CID, DTXSID, etc.)
 - DOI of primary literature
 - Retrieval timestamp + ETag from source server
 - SHA-256 checksum

- License (CC0 / CC-BY / public domain / proprietary-attribution-only)
- FAIR principles: Findable (rich metadata), Accessible (HTTPS, no auth), Interoperable (open formats: mmCIF, SDF, JSON, CSV), Reusable (license + provenance per file).
- Single `(provenance.json)` at the bundle root cross-links every file to its sourcing record.

6c. Python toolkits.

- **Biopython** 1.85+ (`(Bio.PDB)`, `(Bio.PDB.alphafold_db)`, `(Bio.PDB.MMCIFParser)`)
- **gemmi** 0.7+ — fast CIF/mmCIF/BinaryCIF read/write in C++/Python; preferred for archive-scale CIF
- **RDKit** 2026.03.2 — small molecule + polymer cheminformatics
- **OpenBabel** 3.1.1 — format conversion fallback
- **MDAnalysis** 2.x — trajectory + structure I/O
- **Biotite** 1.x — `(biotite.database.rcsb.fetch())` + structure analysis
- **pypdb** — thin RCSB wrapper
- **rcsb-api** (Piehl et al. 2025) — official RCSB Python client
- **pubchempy** 1.0.5 — PubChem PUG-REST
- **rdkit-pypi** + **chembl_webresource_client** for ChEMBL
- **molfeat** (Datamol) — unified molecular featurizers
- **sentence-transformers** for polyBERT embeddings
- **transformers** (HuggingFace) for ChemBERTa-2 / MolFormer

6d. CDN, caching, and asset optimization.

- BinaryCIF: typically 30-70% the size of mmCIF.gz; native to Mol*.
- Serve with `(Cache-Control: public, max-age=31536000, immutable)` for versioned filenames.
- Pre-compress: `(gzip -9)` and `(brotli -q 11)` variants served via Content-Negotiation.
- HTTP/2 multiplexing for the dozens of small JSON files per garment.
- Range requests on big SDF/PDB bundles allow Mol* lazy loading.
- Service Worker caches the static artifacts for offline demo capability.

End-to-End Data-Sourcing Architecture (the rigorous version)

Layer 0 — Source registry (`(provenance.json)`): cryptographically signs every artifact with source DB + accession + retrieval timestamp + DOI + SHA-256 + license. Expert judges can click

any structure in the UI → see its provenance card → click through to rcsb.org / pubchem / alphafold to verify the byte-identical source.

Layer 1 — Experimental anchors (CC0/public domain, frozen):

- 20 PDB entries (UMG-SP family + PET hydrolase analogs + Ser-His-Asp templates) downloaded as .bcif via files.rcsb.org / models.rcsb.org.
- ~30 small molecules (monomers, chain extenders, substrates, hazard chemicals) downloaded as SDF via PUG-REST.
- DSSTox CSV subset for hazard cross-refs.

Layer 2 — Predicted-structure references (CC-BY 4.0, frozen):

- AlphaFold v6 model for UMG-SP2 sequence (for comparison vs the crystal) + sister sequences from ESM Atlas (for design diversification).
- pLDDT / pTM displayed prominently in the UI.

Layer 3 — ML training data + models (mixed licenses, attribution-clean):

- PI1M CSV (~1M PSMILES), figshare DOI.
- polyBERT HF model (kue lumbus / polyBERT) for polymer fingerprinting.
- ChemBERTa-2 / MolFormer-XL for substance-of-concern featurization.
- PoLyInfo subset reference values cited individually (no scraping).

Layer 4 — Engine outputs (clearly tagged "PREDICTED, NOT VALIDATED"):

- Engine A designed elastane variants → PSMILES + BigSMILES + RDKit ETKDgv3 3D conformer + polyBERT-predicted Tg/Tm/elongation/modulus.
- Engine B designed urethanase → predicted PDB (Rfdiffusion2 backbone, LigandMPNN sequence, Boltz-2 + PLACER scoring, FoldSeek nearest-neighbor analysis) — every file labeled with the design pipeline version + scores.
- Engine C Digital Recyclability Passport JSON cross-references back to the molecule registry by CID/DTXSID/PSMILES hash.

Layer 5 — Frontend rendering:

- Mol* loads .bcif by URL for any PDB-anchored protein.
- 3Dmol.js / RDKit.js for small molecules.
- React Three Fiber for the polymer-chain narrative + cleavage animation (zoom from garment → fiber → polymer chain → carbamate bond → enzyme active site).

The exact download list (paste into `make data`):

```
# Engine B – Protein anchors (BinaryCIF preferred)
curl -o proteins/8WDW.bcif https://models.rcsb.org/8WDW.bcif      # UMG-SP2 apo
curl -o proteins/9FZW.bcif https://models.rcsb.org/9FZW.bcif      # UMG-SP2 (2nd
deposit)
curl -o proteins/8S7Z.bcif https://models.rcsb.org/8S7Z.bcif      # UMG-SP1
curl -o proteins/9FVF.bcif https://models.rcsb.org/9FVF.bcif      # UMG-SP3
curl -o proteins/6C6G.bcif https://models.rcsb.org/6C6G.bcif      # AtzE biuret
amidase
curl -o proteins/5H6S.bcif https://models.rcsb.org/5H6S.bcif      # hydrazidase
curl -o proteins/8ES6.bcif https://models.rcsb.org/8ES6.bcif      # ClbL
curl -o proteins/3A2P.bcif https://models.rcsb.org/3A2P.bcif      # NylA
curl -o proteins/6EQE.bcif https://models.rcsb.org/6EQE.bcif      # IsPETase
curl -o proteins/7SH6.bcif https://models.rcsb.org/7SH6.bcif      # FAST-PETase
curl -o proteins/6THS.bcif https://models.rcsb.org/6THS.bcif      # LCC-ICCG S165A
curl -o proteins/7QVH.bcif https://models.rcsb.org/7QVH.bcif      # HotPETase
curl -o proteins/1TCA.bcif https://models.rcsb.org/1TCA.bcif      # CalB lipase
curl -o proteins/40YY.bcif https://models.rcsb.org/40YY.bcif      # HiC cutinase
curl -o proteins/6QZ4.bcif https://models.rcsb.org/6QZ4.bcif      # MHETase

# Engine A – Monomer / chain extender SDFs
for CID in 13242 17394 16156 8064 3301; do
  curl -o molecules/CID_${CID}.sdf \
    "https://pubchem.ncbi.nlm.nih.gov/rest/pug/compound/cid/${CID}/record/SDF?
record_type=3d"
done

# Engine C – Hazard / passport references
curl -o passport/pfas_master.csv \
  https://comptox.epa.gov/dashboard/chemical-lists/PFASMASTER/export
curl -o passport/fashionpedia_ontology.json \
  https://huggingface.co/datasets/detection-
datasets/fashionpedia/resolve/main/ontology.json
# H&M dataset via Kaggle CLI:
kaggle competitions download -c h-and-m-personalized-fashion-recommendations
```

Recommendations

Stage 0 (pre-hackathon prep): Download the ~50-file artifact bundle above into a version-controlled `artifacts/v0.1.0/` directory with SHA-256s. Bundle ships at <15 MB compressed.

Stage 1 (week 1 build): Wire Mol* viewer to load .bcif by accession; build RDKit-based polymer-chain renderer for the segmented PU-urea; pre-compute 3-5 polymer designs from Engine A and 1 enzyme design from Engine B; serialize all to JSON + SDF + PDB.

Stage 2 (demo polish): Add ESM Atlas / AlphaFold v6 comparison view; add CompTox Dashboard hazard look-ups; add Higg MSI + PEFCR v3.1 derived environmental fields to passport.

Thresholds that would change strategy:

- If UMG-SP2 structure 8WDW resolution proves insufficient for active-site templating → switch primary anchor to UMG-SP3 (9FVF) which has a product-bound state.
- If polyBERT 80M-trained embeddings underfit the elastane regime → fine-tune with a PoLyInfo PU subset (~3000 rows).
- If H&M dataset lacks enough children's garments → augment with Fashionpedia + synthetic Carter's-style records.

What is real vs predicted vs synthetic (the honesty section)

Layer	Status	Examples	Disclosure tag
Experimental crystal structures	Real, measured	8WDW (UMG-SP2), 6EQE (IsPETase), 1TCA (CalB)	"Experimental X-ray, RCSB PDB, CC0"
AlphaFold predictions	Predicted	AF-{UniProt}-F1 models	"AlphaFold v6 prediction, pLDDT shown, CC-BY 4.0"
ESM Atlas predictions	Predicted (lower confidence than AF)	Metagenomic ESMFold models	"ESMFold prediction, pLDDT+pTM shown, CC-BY 4.0"
PubChem small molecules	Real (chemistry)	HDI CID 13242	"PubChem, public domain"

Layer	Status	Examples	Disclosure tag
PubChem 3D conformers	Predicted (MMFF)	3D SDF from PUG-REST	"PubChem 3D conformer, MMFF-optimized"
RDKit ETKDgv3 polymer 3D	Predicted (computed)	Engine A polymer chains	"RDKit ETKDgv3 + CSD torsion priors"
PI1M polymers	Synthetic (RNN-generated, in-silico)	1M PSMILES	"PI1M generated, may not be synthesizable"
PoLyInfo property values	Real (measured, peer-reviewed lit)	Tg, Tm, etc.	"PoLyInfo (NIMS), cited per row"
polyBERT predictions	Predicted (ML)	Tg/Tm/elongation for Engine A designs	"polyBERT ML prediction, $\pm\sigma$ shown"
Engine B designed enzyme	Predicted, NOT validated	RFdiffusion2 $\rightarrow$ Boltz-2 outputs	"REWEAR-FUSED design, computational only"
H&M product records	Real (commercial catalog)	article_id-keyed	"H&M Kaggle, research use"
Carter's-style baby garment records	Synthetic + augmented	onesie/sleeper/legging composition	"Synthetic, blend ratios from category priors"
Higg MSI scores	Real (industry LCA)	per-material 10-pt average baseline	"Higg MSI v3.11 reference, Cascale"
Engine C passport	Composed (real + predicted)	per-garment	each field tagged with source class

Caveats

1. **UMG-SP family caveat:** although 8WDW was deposited in 2023, full public release came only late 2024; some 2023-era papers reference an AlphaFold2 model based on GenBank OP972510 rather than the crystal. The triad is Ser-cisSer-Lys (amidase-signature), NOT canonical Ser-His-Asp; REWEAR-FUSED's targeted Ser-His-Asp triad must be justified as a designed re-anchoring, not a literal copy of UMG-SP2.

2. **NIMS PoLyInfo** — no public bulk download, no scraping. Use PI1M as the open proxy for ML training; cite PoLyInfo only for individually-fetched reference values.
3. **Higg MSI** is only accessible through Worldly (paid). For the demo, use published Higg MSI normalized scores (10 = average material) from Cascale documentation as reference values; do not claim "live API integration."
4. **DeepFashion** requires a signed agreement for image data; metadata-only use is fine but image redistribution is not.
5. **Ketcher v2+ is Apache-2.0**; older versions had AGPL; check that you're on ≥ 2.0 for commercial-friendly licensing.
6. **CSD** is commercial. Demo should rely on RDKit ETKDGV3 (which embeds CSD-derived torsion preferences) rather than directly querying CSD.
7. **GINETEX care symbols are trademarked**; reproduce only with attribution; for production use, license is required.
8. **PubChem 3D conformers are MMFF-optimized**, not always experimental geometries; for the cleavable elastane substrate, generate fresh ETKDGV3 conformers and document the protocol.
9. **API rate limits**: RCSB ≤ 5 req/s, PubChem ≤ 5 req/s, AlphaFold DB ~ 1 req/s recommended. Pre-fetch all artifacts at build time; never depend on a live API during a judged demo.
10. **MlhB**: no PDB entry was located under that exact name during research; the closest functional analogue with a deposited structure is the paraben amidase (PaAm, see Oxford Acad bbb/zbae083, 2024) and the maleamate amidase Ami (4L08). If MlhB is needed specifically, fall back to an AlphaFold v6 prediction of the published sequence.
11. **UMG-SP3 inhibitor- and product-bound depositions**: Rotilio 2025 (Angew Chem 64, e202419535) describes three states for UMG-SP3 — ligand-free (PDB 9FVF), suicide-inhibitor-bound, and product-bound. The two ligand-bound entries are companions to 9FVF (likely 9FVG and 9FVH by deposition sequence); confirm by direct lookup on RCSB before relying on them. [Wiley Online Library](#)